

EBN122 EKSAMEN/EXAMINATION (NOVEMBER 2016)

ANTWOORDBLAD / ANSWER SHEET

Surname / Van:	Memo.	Student #:								
Initials/ Voorletters:										

Q1	18	Q2	32	Q3	20	Q4	24	Tot:	/90
----	----	----	----	----	----	----	----	------	-----

QUESTION 1 / VRAAG 1 (18)		
1.1	a) $p(35s) = +13.125 \text{ W}$ ✓	b) $Q = 40 \text{ C}$ ✓ ✓
1.2	$I_1 = 0.06 \text{ A}$ ✓	$V_2 = -1.2 \text{ V}$ ✓ ✓
1.3	$R = 16 \Omega$ ✓	
1.4	$R_{cd} = 155 \Omega$ ✓	
1.5	$I_s = -0.55 \text{ A}$ ✓ ✓	$V_1 = 1.875 \text{ V}$ ✓ ✓

QUESTION 2 / VRAAG 2 (32)				
2.1	$a = 18$ ✓	$b = 6$ ✓	$c = -10$ ✓	$d = 9$ ✓
2.2	$e = -1$ ✓	$f = 1$ ✓	$g = 5$ ✓	$h = 28$ ✓
2.3	$V_Q = -16 \text{ V}$ ✓ ✓			
2.4	a) $I_B = 0.1 \text{ mA}$ ✓ ✓	b) $V_{CE} = 11.58 \text{ V}$ ✓ ✓		
2.5	a) $V_{O(1A)} = -20 \text{ V}$ ✓ ✓	b) $V_{O(30V)} = 60 \text{ V}$ ✓ ✓		
2.6	a) $R_T = 6 \text{ k}\Omega$ ✓ ✓	b) $V_X = 10 \text{ V}$ ✓ ✓		
2.7	a) $V_{TH} = 7.5 \text{ V}$ ✓ ✓	b) $R_{TH} = 15 \Omega$ ✓ ✓		
2.8	a) $I_N = -1.5 \text{ A}$ ✓ ✓	b) $R_N = 40 \Omega$ ✓ ✓		
2.9	$V_{TH} = \pm 12 \text{ V}$ ✓ ✓			

QUESTION 3 / VRAAG 3 (20)			
3.1	$i_0 = 5A$ ✓		
3.2	a) $V_0 = -16V$ ✓	b) $V_0 = -39V$ ✓	
3.3	a) $V_m = 6V$ ✓	b) $V_n = 2V$ ✓	c) $V_o = 6V$ ✓
3.4	$v_C(t) = 8.25V$ ✓		
3.5	$C_{eq} = 10F$ ✓		
3.6	$\omega_C = 0J$ ✓		$\omega_L = 9J$ ✓

VRAAG 4 – QUESTION 4 (24)			
4.1	a) $\bar{V} = 3.54 + j17.68 V$ ✓ (rectangular)		
	b) $v_C(t) = 2 \cos(10^6 t - 155^\circ) V$ ✓		
	c) $R = 6.5\Omega$ or $L =$ ✓ or $C =$		
4.2	a) $\bar{I}_1 = 2.5 \angle 8.13^\circ A$ ✓ (polar)		
	b) $\bar{V}_1 = 0.5 \angle 90^\circ$ ✓ (polar)		
4.3	$\bar{Z}_{ab} = 40 - j40\Omega$ ✓ (rectangular)		
4.4	$a_1 + jb_1 = 1$ ✓	$a_2 + jb_2 = -1$ ✓	
	$a_3 + jb_3 = -2$ ✓	$a_4 + jb_4 = 2 + j$ ✓	
4.5	$\bar{V}_{Th} = 1.414 \angle 0^\circ V$ ✓ (polar)	$\bar{Z}_{Th} = 3.54 \angle -45^\circ \Omega$ ✓ (polar)	
4.6	$\bar{I}_N = 7.07 \angle -45^\circ A$ ✓ (polar)	$\bar{Z}_N = 7.07 \angle -45^\circ \Omega$ ✓ (polar)	